EEG Web Application Interface for Native Integration with Neuroadaptive VR Platforms







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Goal & Vision

Researchers often need to utilize Lab Stream Layer (LSL) for recording and synchronizing EEG data and stimulus event data, and often need to keep such records in .xdf file format. While some programs exist to enable researchers to do this, no convenient web applications appear to streamline the **entire process** of:







1) visualizing the streamed EEG data;
2) allowing researchers to see how common filters and pre-processing methods affect their data in real time;
3) logging and accessing subject data, records, and notes;
4) configuring recording sessions to automatically subscribe to specified EEG and VR data streams;
5) saving all pre-configured LSL data stream names;
6) exporting *fully synced* .xdf files in software-managed directories

Production Roadmap

- ☐ Data Stream Page for Visualizing EEG Data Streams
  - ☑ Functions for Real-Time Pre-Processing of EEG Data
  - ☑ Update All Trial Metadata via Experimental Setup Configuration
  - ☑ Recording Behavior to Save Trial Data Once Session Completes
  - ☐ Plotly Graph to Properly Display All EEG Data in Stream



- ☑ Subjects Page for Managing Subject Data Used for Trials
  - ☑ Save All Input Subject Data to Local SQL Database

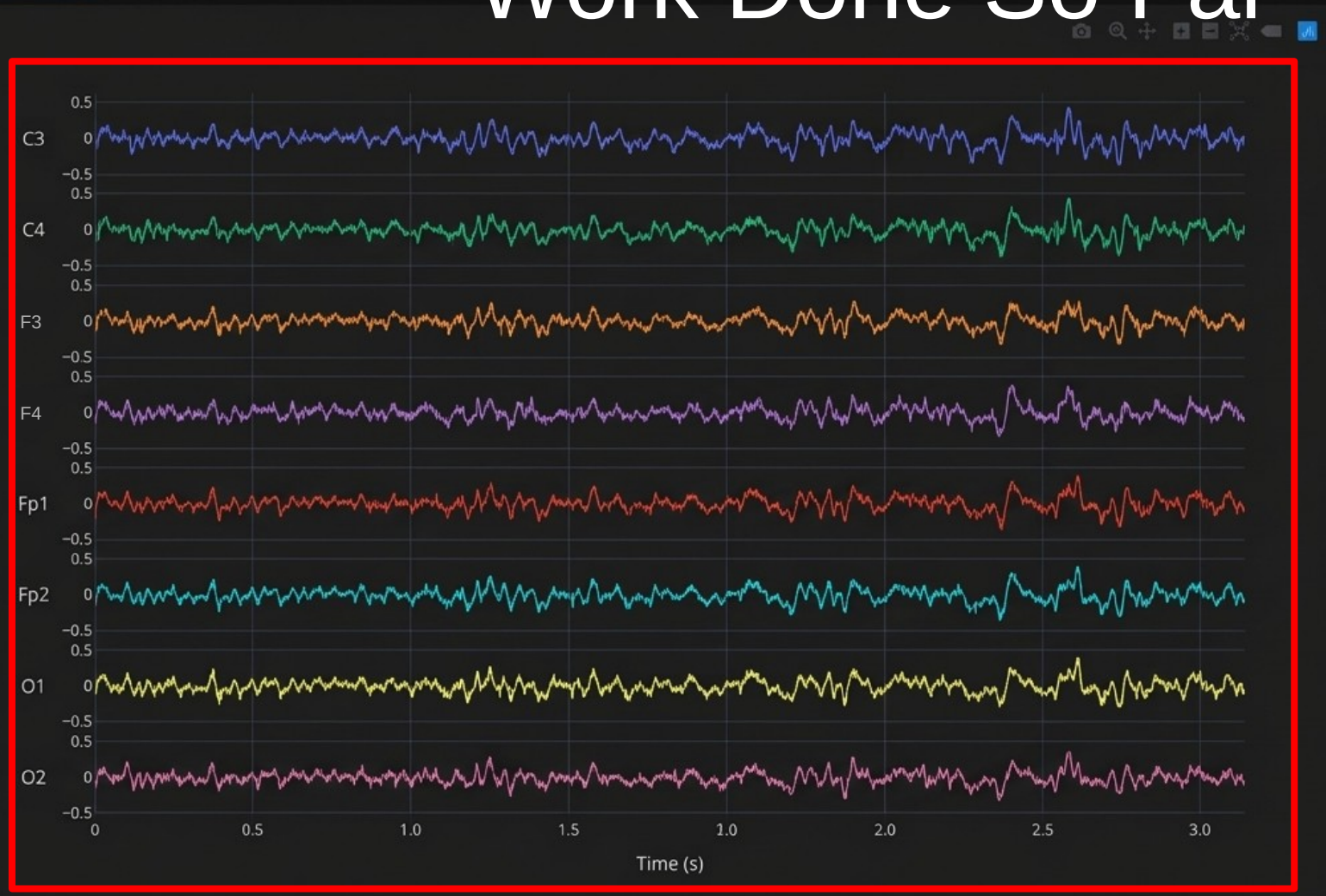


- ☑ Experimental Setup Page for Configuring Trial Data & Metadata
  - ☑ Refresh All Default and Dropdown Values via Local SQL Database
  - ☑ Create Functionality for Adding New Entries Saved to Local SQL Database

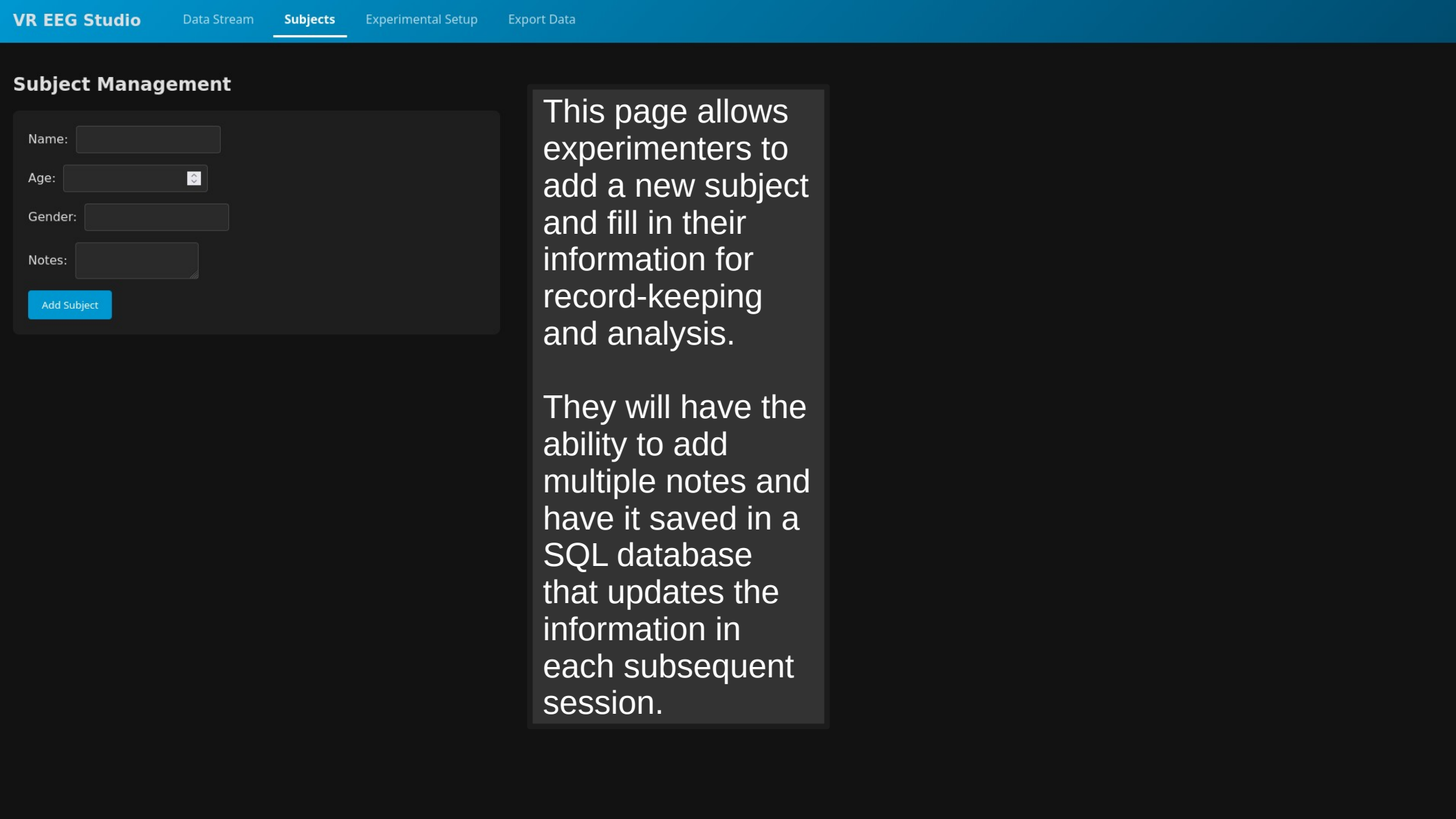


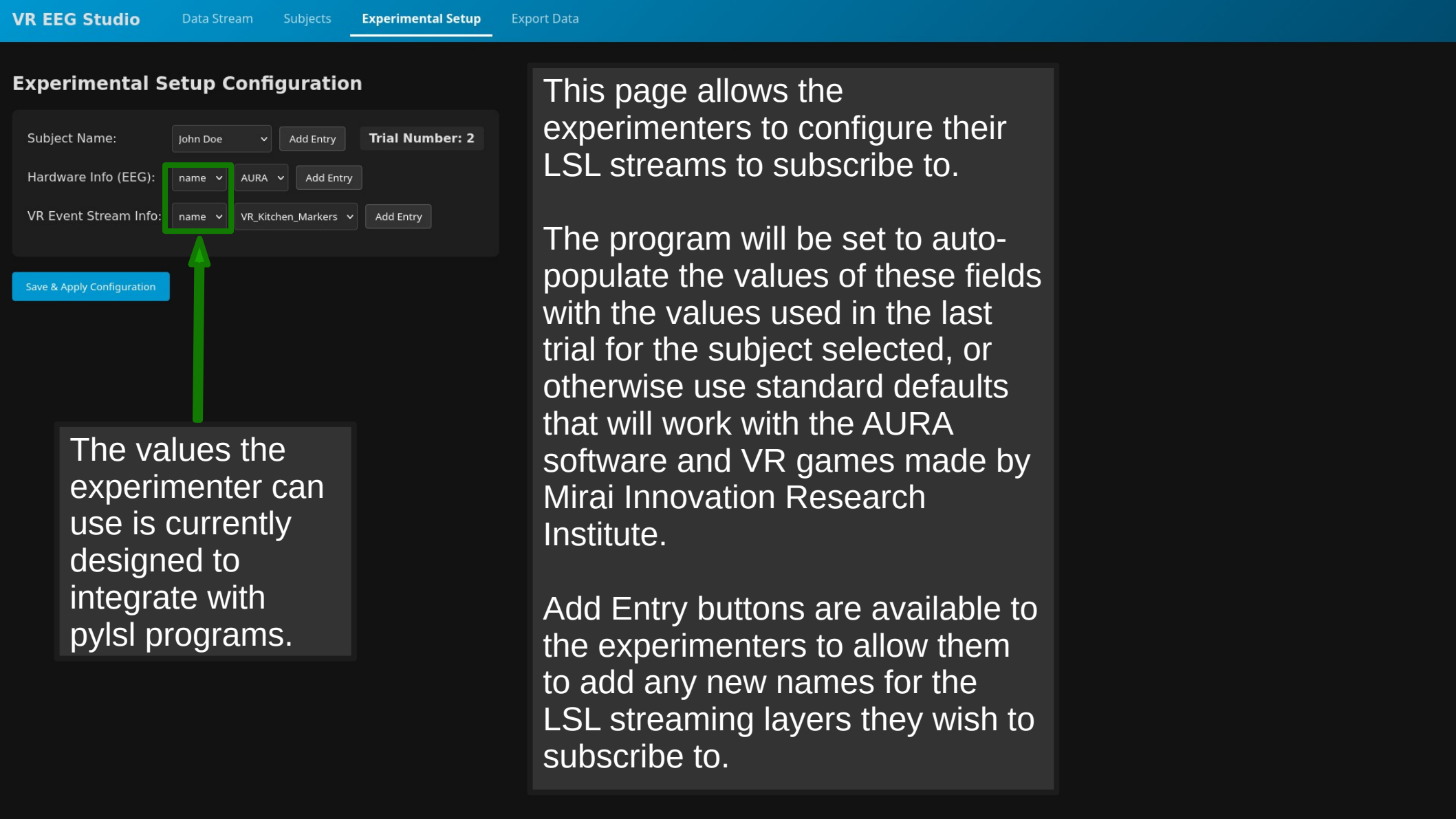
- ☐ Export Data Page
  - ☐ Sync and Manage Data Streams through LSL's LabRecorder Software CLI
  - ☑ Allow Experimenter to Export Data to .xdf Files *After Finished Trials*

Work Done So Far



This page serves to show the experimenter the channels of the EEG data stream, allow the experimenter to apply filters in real time, and record a trial of data right from the web app.


VR EEG Studio
Data Stream
Subjects
Experimental Setup
Export Data
Subject Management
Name:
Age:
Gender:
Notes:
Add Subject
This page allows experimenters to add a new subject and fill in their information for record-keeping and analysis.
They will have the ability to add multiple notes and have it saved in a SQL database that updates the information in each subsequent session.


VR EEG Studio
Data Stream
Subjects
Experimental Setup
Export Data
Experimental Setup Configuration
Subject Name:
John Doe
Add Entry
Trial Number: 2
Hardware Info (EEG):
name
AURA
Add Entry
VR Event Stream Info:
name
VR_Kitchen_Markers
Add Entry
Save & Apply Configuration
The values the experimenter can use is currently designed to integrate with pylsl programs.
This page allows the experimenters to configure their LSL streams to subscribe to.
The program will be set to auto-populate the values of these fields with the values used in the last trial for the subject selected, or otherwise use standard defaults that will work with the AURA software and VR games made by Mirai Innovation Research Institute.
Add Entry buttons are available to the experimenters to allow them to add any new names for the LSL streaming layers they wish to subscribe to.

Export Session Recordings

Export recorded session streams out to local XDF format files.

Export EEG Data | Export All Data

This page allows the experimenter to export only the EEG data or all synced data to .xdf files *only when the experimenter has finished a new trial*. These files save to a local directory managed by the program.

Next Steps In Completion

1) Fully Integrate Pre-Processing Methods Into Real-Time Data



2) Fix Bugs with Display of Real-Time Data in Plotly Graph



3) Finish Integration of LabRecorder Software for Syncing Data

4) Verify Full Run of Software Using AURA and VR Game



Validation Methodology & Setup

- This software will be validated using AURA software and a VR neuroadaptive game which already has stimulus event markers that are published via the Lab Streaming Layer.



- The software's ability to show EEG data streams in real-time with all combinations of five different versions of pre-processing, ability to manage and update trial data via the SQL database, and ability to export properly synced data to .xdf files will be validated at the end of production.



- The setup will be made to be as streamlined as possible, allowing the end user to either run a single command in a terminal or run a single software executable that will open the program and have default configurations automatically set up to work with AURA software and VR neuroadaptive games made by Mirai Innovation Research Institute.

